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IC DESIGN INTRODUCTION

Lab 2: CMOS Inverter design – Circuit to Layout

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1. Abstract

This report provides a comprehensive theoretical overview from Circuit to Layout of the Complementary Metal-Oxide-Semiconductor (CMOS) inverter, a fundamental component in digital logic design. The analysis explores the structural operation of the inverter, its switching characteristics under varying input conditions, and the critical design principles—specifically transistor sizing—required to achieve symmetric and balanced performance.

2. Introduction

The CMOS inverter is the most basic logic gate in modern digital integrated circuits. Its primary function is to output the logical opposite of its input. Understanding the operational mechanics, switching conditions, and sizing rules of the CMOS inverter is essential for designing robust, low-power, and high-speed digital systems.

3. CMOS Inverter Operation

A standard CMOS inverter is constructed using two types of enhancement-mode MOSFETs: one PMOS transistor and one NMOS transistor.

- Pull-Up Network (PUN): The PMOS transistor is connected to the positive supply voltage (VDD) and acts as the pull-up network.
- Pull-Down Network (PDN): The NMOS transistor is connected to the ground (GND) and acts as the pull-down network.

The output voltage of the inverter is strictly dependent on the input voltage applied simultaneously to the gates of both transistors.

3.1. Operational States

Logic Low Input (0 V**):** When the input voltage is low, the PMOS transistor is turned ON, establishing a conducting path to VDD. Simultaneously, the NMOS transistor is turned OFF. This configuration pulls the output voltage high to VDD.

Logic High Input (VDD**):** Conversely, when the input voltage is high, the PMOS transistor turns OFF, disconnecting the path to VDD. The NMOS transistor turns ON, creating a conducting path to ground. This pulls the output voltage low to GND.

Truth Table:

Input	Output
0 (GND)	1 (VDD)
1 (VDD)	0 (GND)

3.2. Switching Condition:

The exact switching point (or inversion point) of the CMOS inverter occurs when the drain current of the NMOS transistor I_{Dn} perfectly matches the drain current of the PMOS transistor I_{Dp}.

$$I_{Dn} = I_{Dp}$$

At this specific operational point, the input voltage is equal to the output voltage. This value is defined as the switching threshold voltage of the inverter:

$$V_{in} = V_{out} = V_{TH}$$

The threshold voltage is a critical parameter because it dictates the noise margin and the exact point at which the logic state is considered flipped.

4. CMOS Design Principles: Transistor Sizing

To design a highly efficient and reliable CMOS inverter, one must account for the physical properties of the charge carriers within the semiconductor material.

4.1. Carrier Mobility Discrepancy

In silicon, the charge carriers for NMOS transistors are electrons, while the charge carriers for PMOS transistors are holes. A fundamental physical property of silicon is that the mobility of electrons μ_n is significantly higher than the mobility of holes μ_p.

$$\mu_n > \mu_p$$

4.2. Sizing Relationship

Because holes move slower than electrons, a PMOS transistor will inherently have less drive strength (conduct less current) than an identically sized NMOS transistor. If both transistors were drawn with the exact same physical dimensions, the inverter's transition from low-to-high would be slower than its transition from high-to-low, resulting in asymmetric switching behavior.

To compensate for this physical discrepancy and ensure symmetric switching characteristics (equal rise and fall times), the PMOS transistor must be made physically larger (wider) than the NMOS transistor. The geometric width (W) sizing relationship is directly proportional to the ratio of their carrier mobilities:

$$\frac{W_p}{W_n} \approx \frac{\mu_n}{\mu_p}$$

By adhering to this sizing relationship, designers ensure that the pull-up and pull-down networks possess equivalent current-driving capabilities, thereby achieving a balanced, ideal switching threshold:

$$V_{TH} \approx V_{DD}/2$$

4. Tools / Software

- Cadence Virtuoso and Spectre are used to design, simulate, and verify the CMOS inverter circuit and layout.

5. Lab Procedure

Part A: Schematic Design

In this part, a CMOS inverter schematic is created using one PMOS transistor connected to VDD and one NMOS transistor connected to ground. The gates of both transistors are connected together to form the input, while the drains are connected together to form the output.

- Create CMOS inverter
- 1 PMOS (120/100) (top, connected to V_{DD})
- 1 NMOS (120/100) (bottom, connected to GND)
- Common gate = input
- Common drain = output

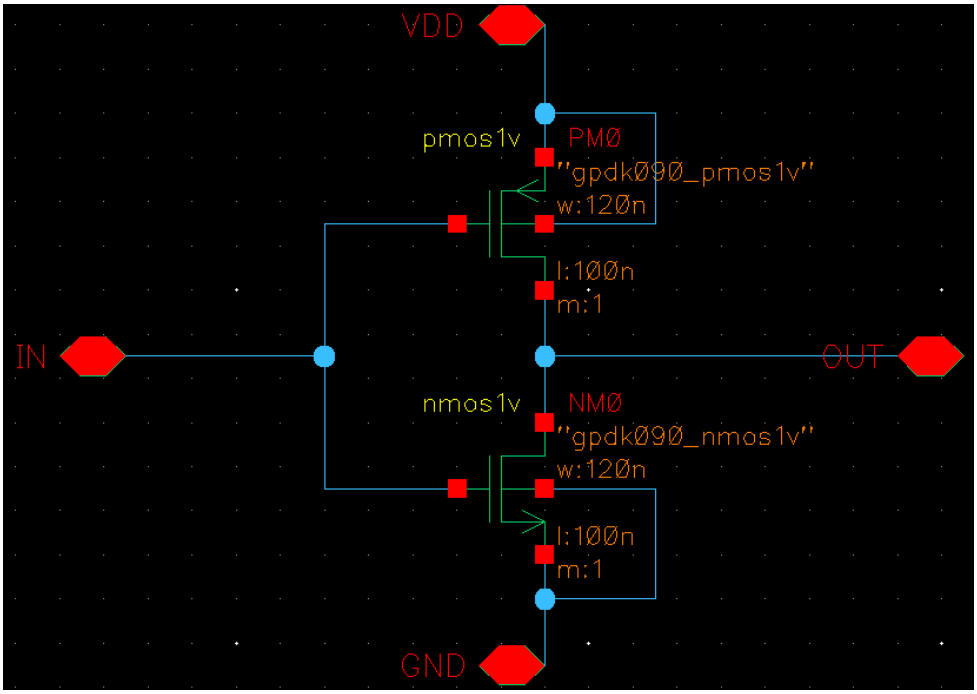


Figure 1: CMOS Inverter schematic

Part B: DC Simulation (VTC)

In this part, the input voltage is swept from 0 V to VDD. The output voltage is plotted against the input voltage to obtain the Voltage Transfer Characteristic (VTC) curve of the CMOS inverter.

- Sweep input: Vin from 0 to VDD
- Plot Vout vs Vin

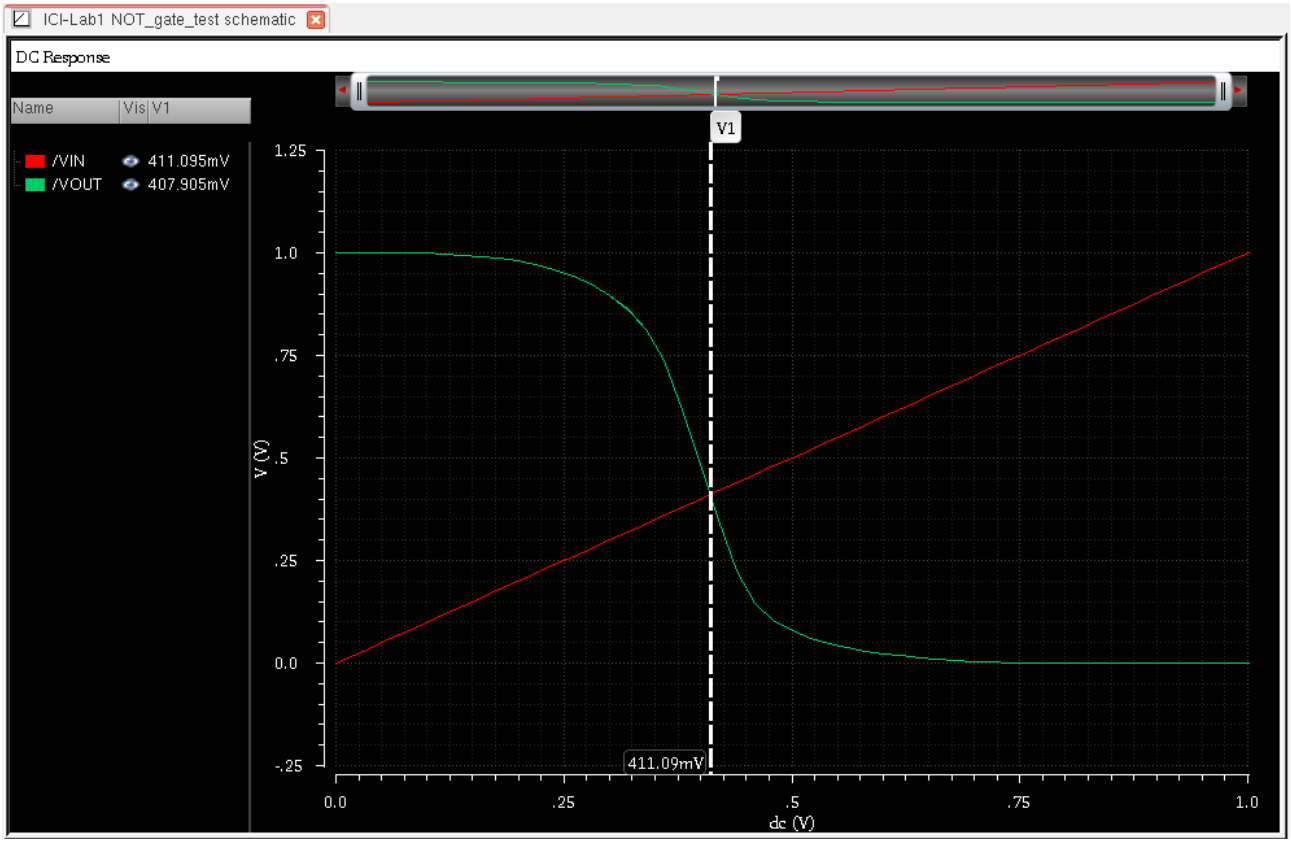


Figure 2: VTC curve of CMOS Inverter

Analysis Remark: $V_{TH} \approx 411\ mV < V_{TH,ideal} = 0.5\ V$

However, for a perfectly symmetric inverter operating at VDD = 1.0 V, the ideal switching threshold is exactly in the middle: $V_{TH} = 0.5\ V$

Because the measured threshold (0.41 V) is significantly lower than the ideal threshold (0.5 V), the VTC curve is skewed to the left.

This visual skew perfectly illustrates the mobility discrepancy discussed in theoretical background:

- The curve shifts left because the NMOS transistor (pull-down) is physically "stronger" than the PMOS transistor (pull-up) at equal gate voltages. As V_{in} rises, the NMOS begins to overpower the PMOS too early, dragging the output down before V_{in} even reaches 0.5 V.
- Conclusion**:** This schematic uses un-sized or equally sized transistors ($W_p = W_n$). Because electron mobility μ_n is roughly 2.5 times higher than hole mobility μ_p , an equally sized NMOS will dominate.

This result serves as an excellent practical example of why un-sized inverters fail to provide optimal noise margins. To fix this skew and push $V_{TH} = 0.5$ V, the PMOS width must be increased (we predict the ratio to be 2.5 times) relative to the NMOS width.

Predict: $\mu_n \approx 2.5 \text{ x } \mu_\mathrm{p}$

Part C: Transient Analysis

In this part, a pulse signal is applied to the input of the CMOS inverter. The propagation delay, rise time, and fall time are measured from the transient response waveform.

- Apply pulse input
- Measure: Propagation delay (tp), Rise time (tr) and Fall time (tf).

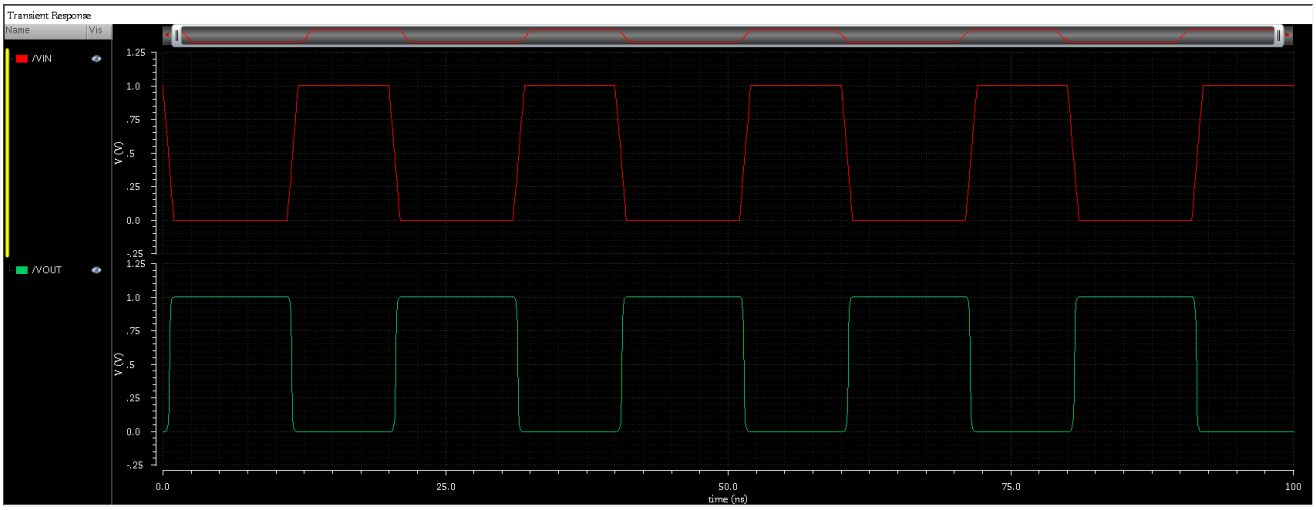


Figure 3: Transient response of CMOS Inverter

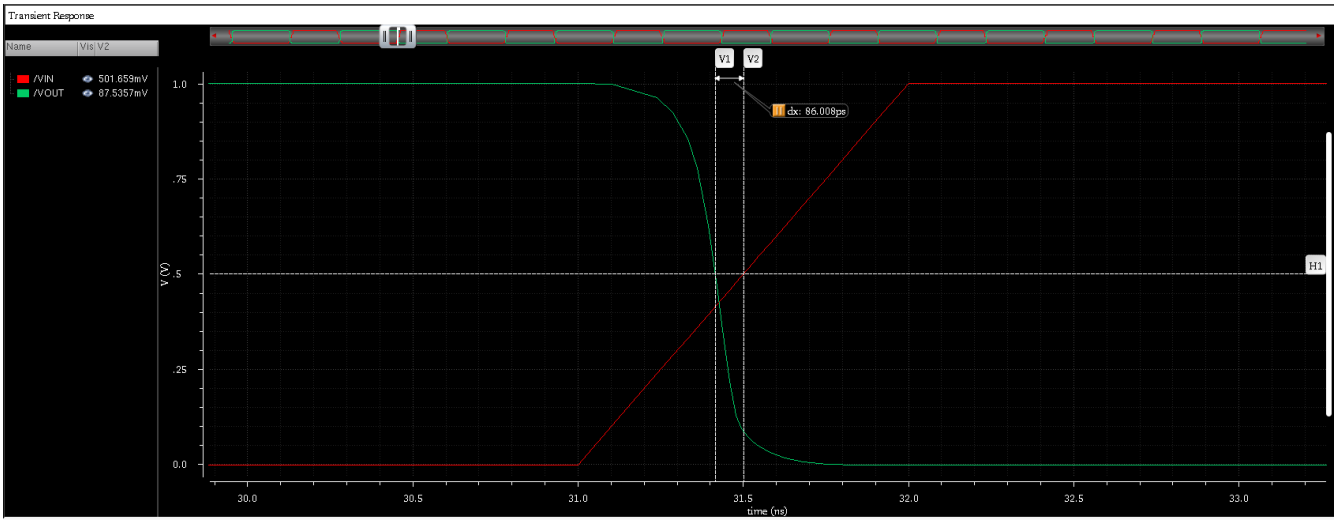


Figure 4: High to Low measurement: $\boldsymbol{T_{H2L}} \approx 86$ ps

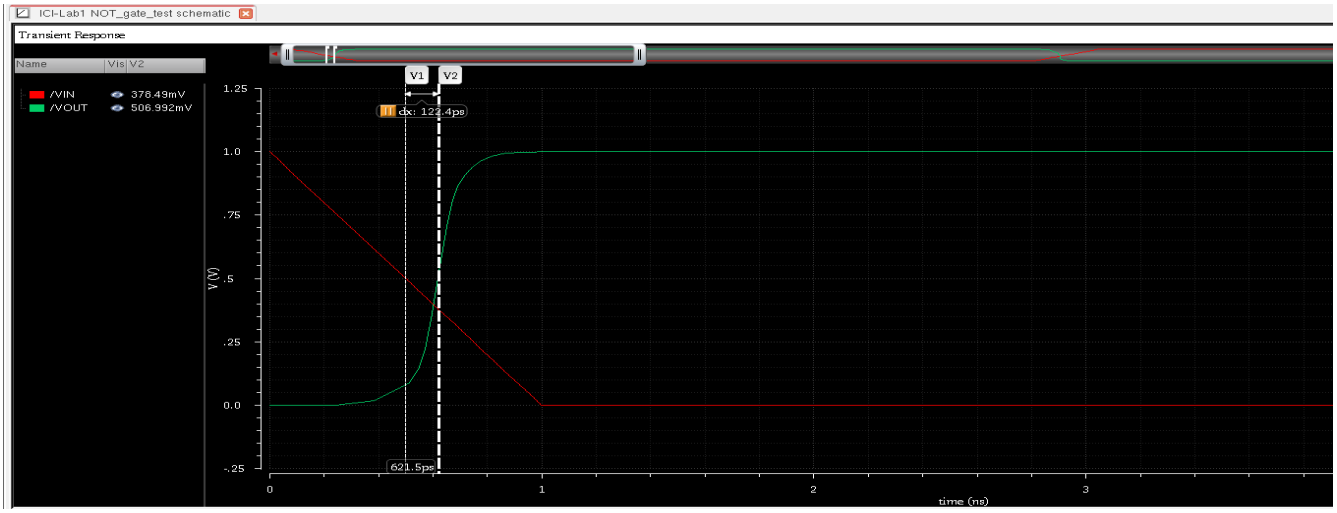


Figure 5: High to Low measurement: $\boldsymbol{T_{L2H}} \approx 122$ ps

<u>Analysis Remark:</u> $\boldsymbol{T_P = (\, T_{L2H} + T_{H2L}\,) / 2 \approx 104}$ **ps.**

The transient analysis results yielding a low-to-high propagation delay T_{L2H} of 122 ps and a high-to-low propagation delay T_{H2L} of 86 ps for an equally sized CMOS inverter ($W_p = W_n$) are entirely consistent with semiconductor physics. These empirical values accurately reflect the inherent physical characteristics of the underlying transistor materials.

1. Physical Justification: Carrier Mobility Discrepancy

The fundamental cause of the observed delay asymmetry lies in the differing carrier mobilities within the silicon substrate:

- T_{H2L} = 86 ps (shorter T_{H2L}) : The discharge of the load capacitance (C_{load}) is facilitated by the NMOS transistor. The majority charge carriers in NMOS devices are electrons, which possess a relatively high mobility. Consequently, the equivalent ON-resistance of the NMOS ($R_{eq,n}$)is comparatively low, resulting in a rapid discharge process and a short T_{H2L}.
- T_{L2H} = 122 ps (longer T_{L2H})**:** Conversely, the charging of the load capacitance is governed by the PMOS transistor. The majority carriers in PMOS devices are holes, which possess significantly lower mobility than electrons. This results in a higher equivalent ON-resistance ($R_{eq,p}$)for a PMOS transistor of identical dimensions, leading to a slower charging process and a correspondingly longer T_{L2H}.

2. Extraction of Technology-Specific Parameters

In digital IC design, transient metrics from an un-sized inverter can serve as a diagnostic tool to estimate underlying technology parameters. Because propagation delay is directly proportional to the equivalent resistance ($T_P \propto R_{eq} \,.\, C_{load}$) the ratio of the delays provides a direct approximation of the mobility discrepancy between electrons and holes for the specific technology node being evaluated.

By calculating the asymmetry ratio, we can quantify this physical property:

$$\text{Asymmetry Ratio} = \frac{T_{L2H}}{T_{H2L}} = \frac{122}{86} \approx 1.418$$

This empirical ratio indicates that, within the simulated technology library, electron mobility is approximately 1.418 times greater than hole mobility ($\mu_n \approx 1.418\ \mu_p$). While traditional bulk CMOS technologies often exhibited ratios closer to 2.0 or 3.0, a ratio of 1.418 is highly characteristic of modern process nodes that employ mobility enhancement techniques, such as strained silicon, to improve PMOS performance

3. Design Optimization

For robust digital systems, achieving a symmetric delay ($T_{L2H} \approx T_{H2L}$) is crucial to maximize operational bandwidth, minimize noise sensitivity, and prevent pulse-width distortion.

Given the empirical evidence that the PMOS network is inherently slower, the fundamental principle of CMOS sizing dictates that the geometric width of the PMOS must be proportionally increased to compensate for the lower hole mobility. The optimal sizing ratio can be directly derived from the calculated delay ratio:

$$W_p \approx 1.418 \text{ x } W_n$$

By scaling the PMOS width to be approximately 1.418 times larger than the NMOS width, the drive currents will equalize ($I_{Dn} \approx I_{Dp}$), the equivalent resistances will match ($R_{eq,n} \approx R_{eq,\text{p}}$), and the inverter will achieve optimal, symmetric switching characteristics ($T_{L2H} \approx T_{H2L}$).

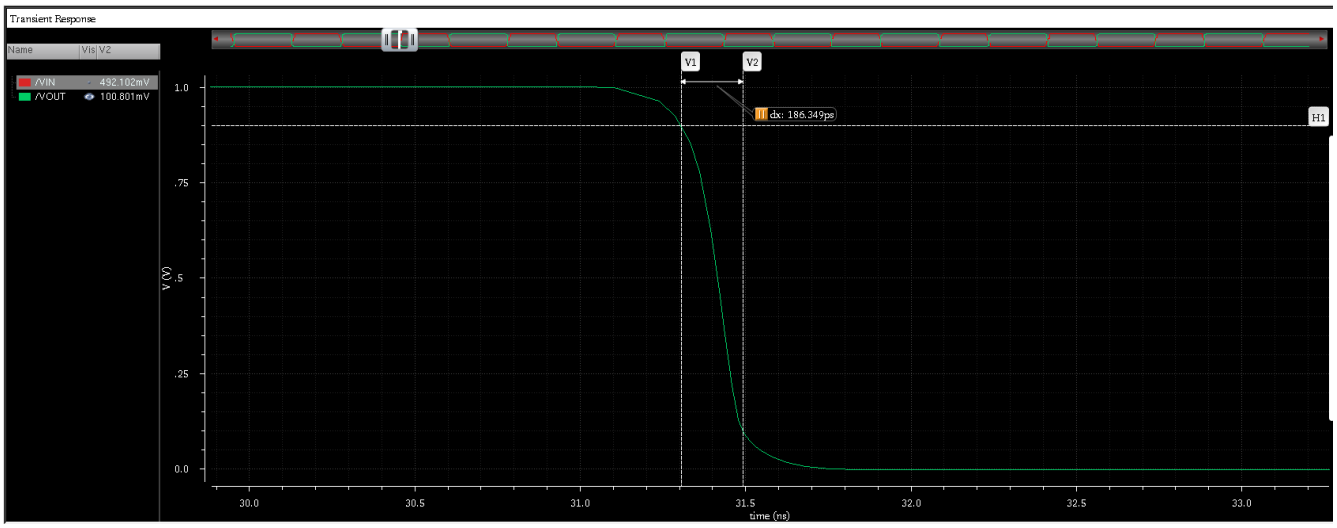


Figure 5: Fall time measurement: $T_f \approx 186$ ps

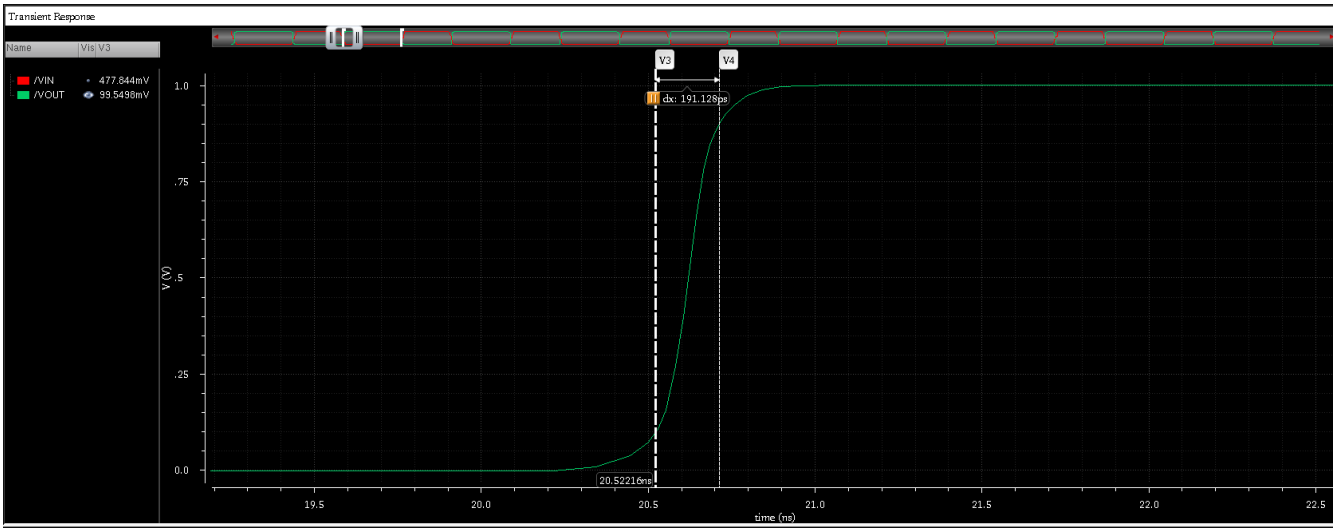


Figure 6: Rise time measurement: $T_r \approx 191$ ps

Remark Analysis: Rise time and Fall time.

1. Phenomenological Consistency: Mobility Discrepancy

At a baseline level, the result $T_r > T_f$ is entirely consistent with the physical analysis established in the previous propagation delay analysis.

- Fall Time**:** Driven by the NMOS discharging the load capacitance to ground. The higher mobility of electrons (μ_n) allows for a faster discharge (186 ps).
- Rise Time**:** Driven by the PMOS charging the load capacitance to VDD. The lower mobility of holes (μ_p) results in a slower charge rate (191 ps).

2. The Velocity Saturation Effect

A critical observation from the result analysis is the severe discrepancy between the delay ratio and the transition time ratio.

- Propagation Delay Ratio: $\frac{T_{L2H}}{T_{H2L}} = \frac{122}{86} \approx 1.418$
- Transition Time Ratio: $\frac{T_r}{T_f} = \frac{191}{186} \approx 1.027$

While the PMOS takes 41.8% longer to reach the halfway point of VDD (50% of VDD), it only takes 2.7% longer to complete the full transition (from 10% to 90%). This compression of the transition ratio is a characteristic of deep submicron (short-channel) CMOS technologies and is primarily caused by velocity saturation.

In highly scaled transistors, charge carriers do not accelerate indefinitely with increasing electric fields. They eventually reach a maximum saturation velocity.

- Because electrons have high mobility, they reach their saturation velocity at relatively low electric fields. Consequently, the NMOS current drive is severely truncated during large voltage swings.
- Holes have lower mobility and thus require much higher electric fields to reach velocity saturation.

Therefore, over the full 10% to 90% voltage swing and vice versa, the NMOS loses its mobility advantage because its electrons are velocity saturated for a significant portion of the transition. This allows the slower PMOS to catch up over the entire duration of the rise time, resulting in remarkably close T_r and T_f values.

4. Design Implications: Short-Circuit Power

While the transition times are surprisingly balanced (191 ps vs 186 ps), their absolute values are relatively large compared to the propagation delays. Slow edge rates (large T_r and T_f) mean that during a switching event, the inverter spends a longer duration in the intermediate voltage zone where both the PMOS and NMOS are partially conducting.

This simultaneous conduction creates a direct path from VDD to ground, leading to short-circuit power dissipation. By applying the sizing principles established previously ($W_p \approx 1.418$ x W_n) we expect to not only equalize the propagation delays but also increase the total drive strength of the pull-up network, thereby sharpening the rise time, reducing short-circuit currents, and improving the overall dynamic power efficiency of the design.

Part D: Layout Design

- Size NMOS : 120/100
- Size PMOS: 120/100
- Gpdk090

Step 1: Device placement

The PMOS transistor is placed inside an n-well, while the NMOS transistor is placed in the p-substrate. The devices are aligned vertically to achieve a compact layout structure.

- PMOS in n-well
- NMOS in p-substrate
- Align vertically for compact layout

Step 2: Layout construction

Polysilicon is used to create the input connection. Diffusion sharing is used for the output node to reduce layout area. Metal routing is implemented for the VDD rail at the top, the GND rail at the bottom, and the output connection in the middle.

- Poly: input connection

- Diffusion sharing for output node
- Metal routing: VDD (top rail), GND (bottom rail) and Output (middle)

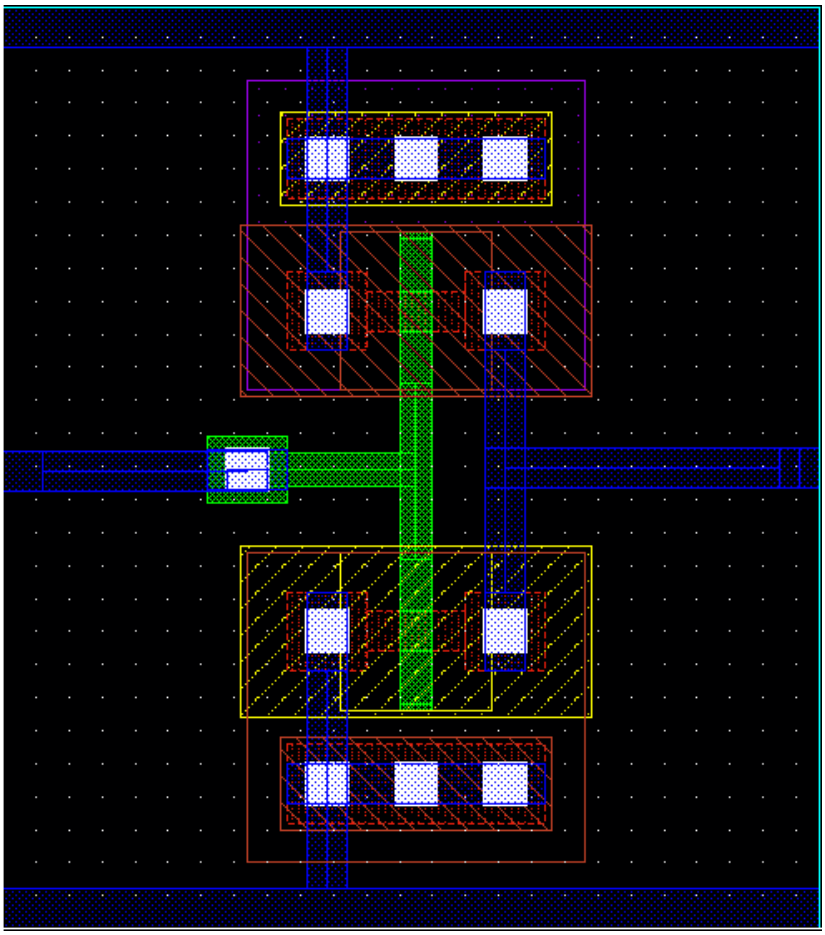


Figure 7: Layout of CMOS Inverter

Step 3: Design rule compliance

Design Rule Check (DRC) is performed to ensure that the layout follows all fabrication rules without violations. Layout Versus Schematic (LVS) verification is also carried out to confirm that the layout connectivity and transistor sizes match the original schematic.

DRC (Design Rule Check) check: Ensure no rule violations

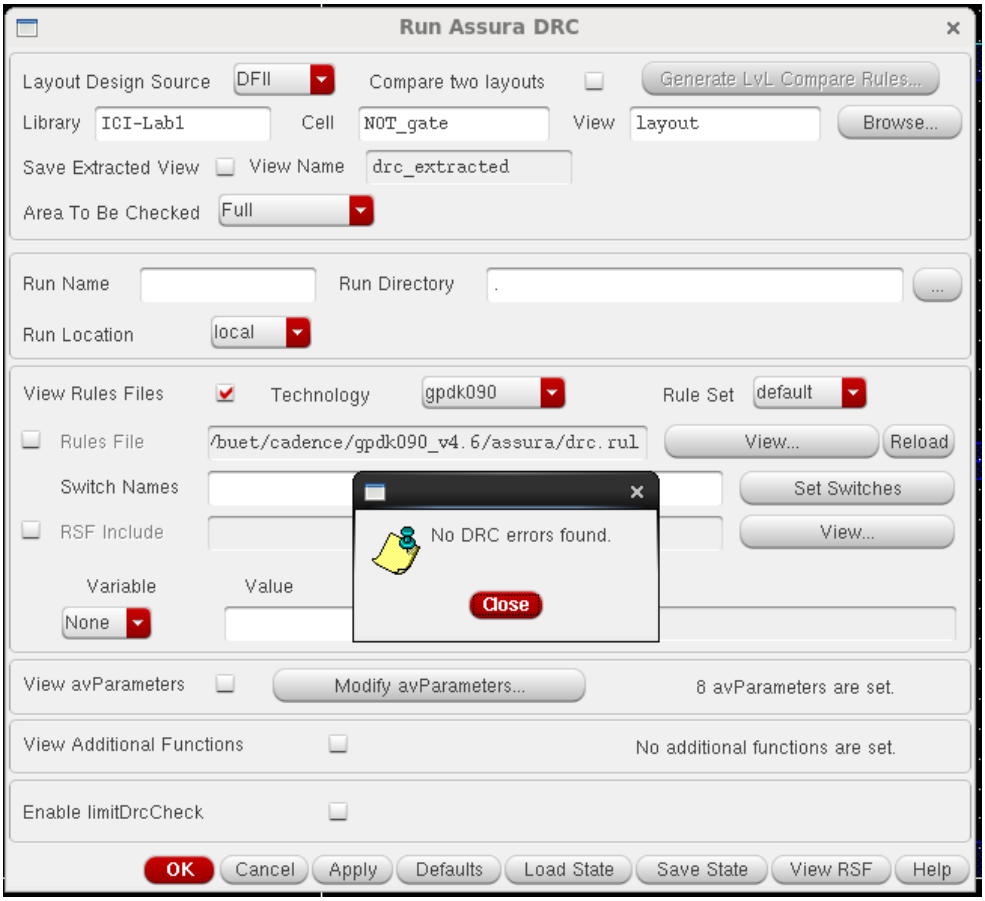


Figure 8: DRC check: pass

LVS (Layout vs Schematic) check: Connectivity and Device sizes match schematic

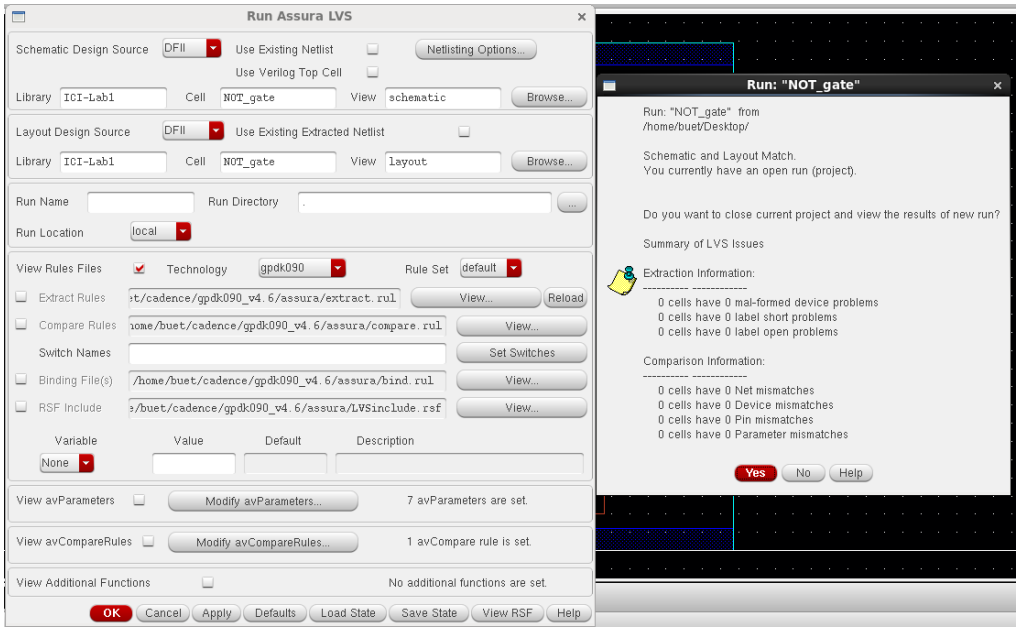


Figure 9: LVS check: pass

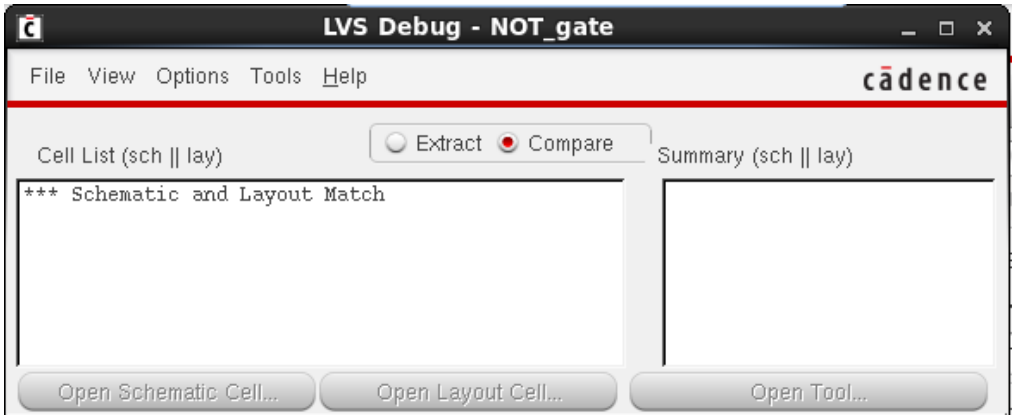


Figure 10: LVS check: pass

RC extraction:

In reality, physical layout creates unintended components (parasitics):

- Every micrometer of metal routing has Resistance (R).
- Every wire running next to another wire, or over the substrate, creates a parallel-plate Capacitor (C).

We need to extract these hidden RC values and back-annotate them into your simulation. Those 122 ps and 86 ps propagation delays we simulated earlier were likely based on idealized schematics. Once running parasitic extraction on the physical layout, we expect that the real-world delays are longer, the rise/fall times are slower, and the power consumption is higher due to the physical burden of driving those actual, physical metal wires.

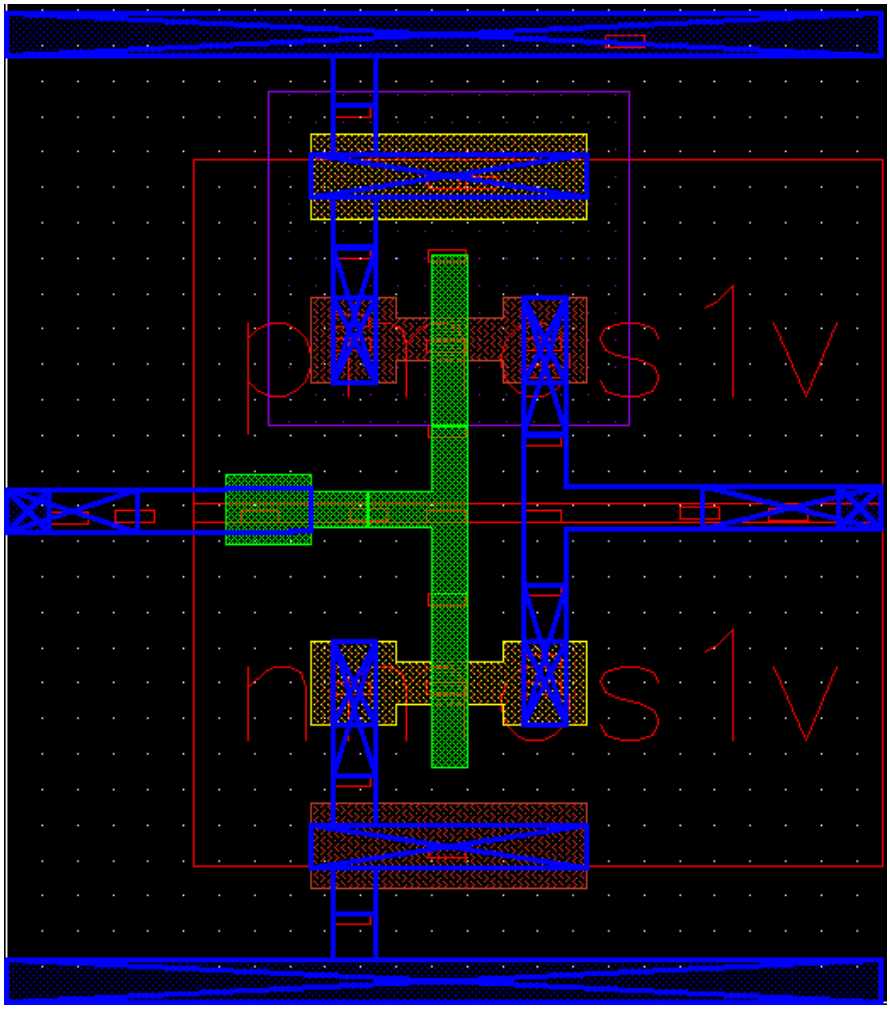


Figure 11: RC parasitic extraction

6. Suggested Self-Practice Activities

To improve understanding of CMOS inverter design, simulation, and layout implementation, students are encouraged to complete the following self-practice exercises:

Exercise 1: PMOS/NMOS Size Optimization

- Change the PMOS width while keeping the NMOS size fixed.
- Observe how the switching threshold voltage changes.
- Determine the PMOS/NMOS size ratio that gives the most symmetric VTC curve.

Exercise 2: Supply Voltage Scaling

- Simulate the CMOS inverter with different supply voltages:
  - 1.2V
  - 1.0 V
  - 0.8 V

- Compare:
    - Switching threshold
    - Propagation delay
    - Rise/fall time
    - Output voltage swing

Exercise 3: Load Capacitance Effect

- Add different capacitive loads at the output node.
- Measure how output load affects:
    - Propagation delay
    - Rise time
    - Fall time
- Explain why larger capacitance slows down the circuit.

7. Assessment Rubric

Criteria	**Description**	**Weight (%)**
Preparation & Theoretical Understanding	Demonstrates understanding of CMOS inverter operation, switching behavior, and CMOS design principles.	10
Schematic Design	Correct implementation of CMOS inverter schematic using NMOS and PMOS transistors. Proper connectivity and device sizing.	15
DC Simulation & VTC Analysis	Correct setup of DC sweep simulation and accurate analysis of the VTC curve and switching threshold voltage.	15
Transient Analysis	Proper pulse input setup and accurate measurement of propagation delay, rise time, and fall time.	15
Layout Design Quality	Clean and compact layout design with proper transistor placement, routing, and layer usage.	15
DRC Verification	Layout passes Design Rule Check without violations.	10
LVS Verification	Layout matches schematic correctly with successful LVS result.	10

Criteria	Description	Weight (%)
Report Quality & Technical Discussion	Clear presentation of simulation results, waveform analysis, screenshots, explanations, and conclusions.	10
Total		100

Performance Guidelines

Excellent (90–100%)

- All schematic, simulation, and layout tasks are completed correctly.
- DRC and LVS pass without errors.
- Strong explanation of CMOS inverter behavior and timing characteristics.
- Well-organized report with clear figures and analysis.

Good (75–89%)

- Most tasks are completed correctly with only minor mistakes.
- Small issues in layout optimization or simulation analysis.
- Good understanding of CMOS inverter operation.

Satisfactory (60–74%)

- Basic functionality works but contains several technical inaccuracies.
- Incomplete explanation of timing or layout verification.
- Minor DRC/LVS issues may remain unresolved.

Unsatisfactory (<60%)

- Incomplete schematic or layout.
- Major simulation/setup errors.
- DRC/LVS failures not corrected.
- Weak understanding of CMOS inverter operation.